SMART RESUME SCREENING TOOL

2510080029-Sudeeptha
2510080037-Latika
2510080043-Sai Siri
2510080061-Kovidha

PROBLEM STATEMENT

- Recruiters receive a large number of resumes for each job opening.
- Manual screening of resumes is time-consuming and inefficient.
- Important skills and qualifications may be overlooked during manual review.
- Comparing resumes with job requirements requires significant effort.
- Delays in screening can slow down the hiring process.
- An automated system is needed to quickly analyze, filter, and rank resumes based on job requirements.

OBJECTIVES

- **Develop an automated resume screening system.**
- **Extract important information from resumes such as:**

1. **Skills**
2. **Education**
3. **Experience**
4. **Certifications**

- **Match candidate profiles with job descriptions.**
- **Rank resumes based on relevance.**
- **Reduce recruiter workload and screening time.**
- **Improve accuracy and fairness in the initial hiring process.**

TECHNOLOGY STACK

Programming Language - Java

Development Tools - Eclipse IDE

Data Structures Used -

- HashMap
- Priority Queue (Heap)

Algorithms Used -

- String Matching (KMP / Rabin-Karp)
- Edit Distance
- Searching Algorithms
- Sorting Algorithms (Merge Sort / Quick Sort)
- Ranking Algorithm

File Handling -

- Java File I/O
- BufferedReader
- FileReader

Database -

MySQL

EXPECTED OUTCOME

The system will:

- **Upload resumes in PDF/Text format.**
- **Extract resume information automatically.**
- **Compare resumes with job descriptions.**
- **Calculate a matching score.**
- **Rank candidates from highest to lowest.**
- **Display the shortlisted resumes for recruiters**

Benefits

- **Faster hiring process**
- **Reduced manual effort**
- **Better candidate selection**
- **Increased efficiency**

PROJECT MODULES:

1. Resume Upload Module

Upload PDF/TEXT resumes.

2. Resume Parsing Module

Extract candidate details.

3. Job Description Module

Enter job requirements.

4. Resume Matching Module

Compare resumes with job descriptions.

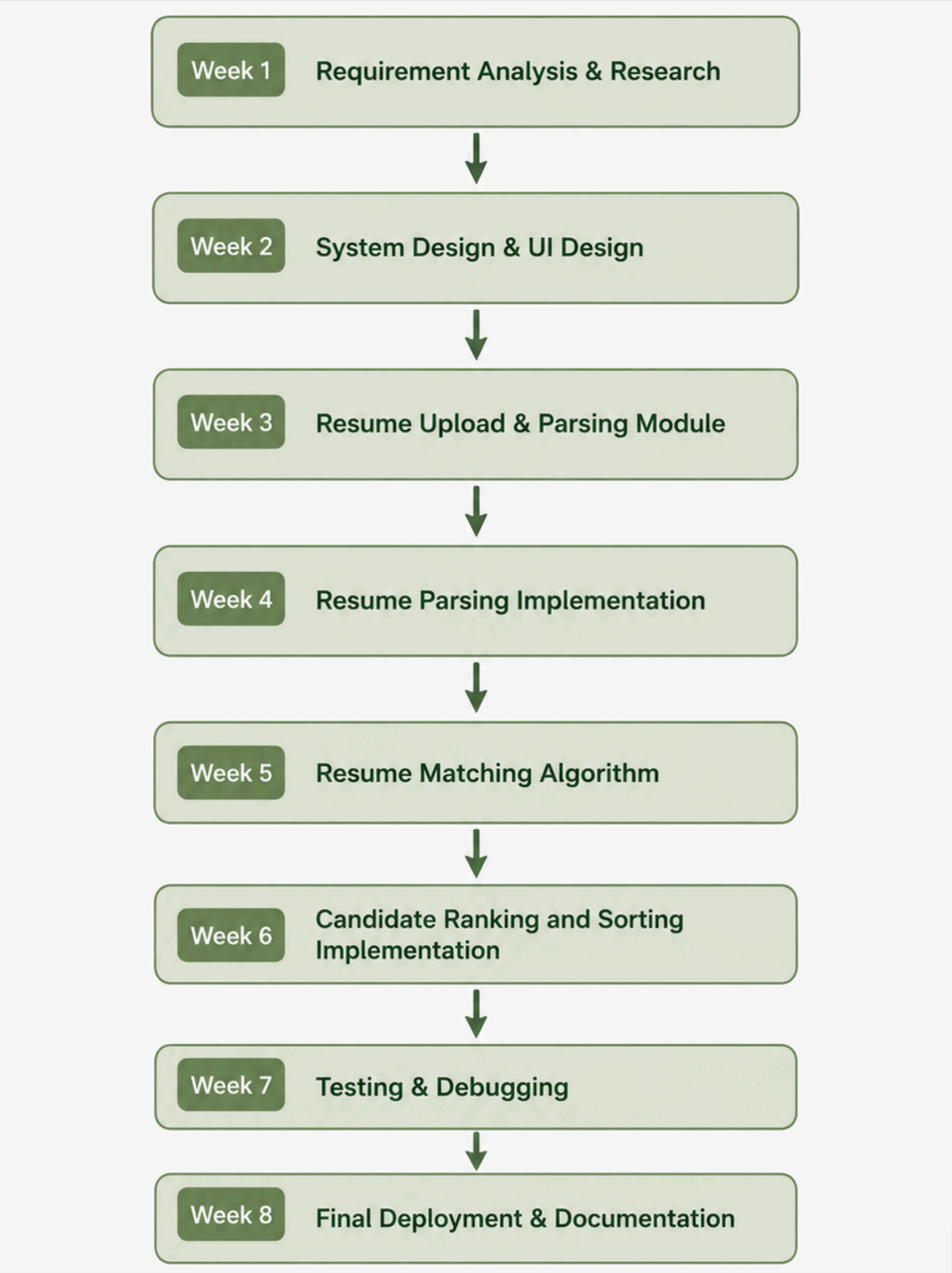
5. Resume Ranking Module

Assign scores and sort resumes.

6. Result Dashboard

Display ranked candidates.

TIMELINE


Week 1
Requirement Analysis & Research
Week 2
System Design & UI Design
Week 3
Resume Upload & Parsing Module
Week 4
Resume Parsing Implementation
Week 5
Resume Matching Algorithm
Week 6
Candidate Ranking and Sorting Implementation
Week 7
Testing & Debugging
Week 8
Final Deployment & Documentation

THE END

THANK YOU